RITIK RAJENDRA DHAGE

MD Candidate | Cardiology Aspirant | Clinical Researcher

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CLINICAL PROFILE

- Dedicated MD Candidate with **5.2+ years of rigorous clinical training**, seeking to leverage a robust foundation in evidence-based medicine into a competitive residency match.
- **Cardiology & Internal Medicine Focus:** Highly driven to advance cardiovascular care through proactive diagnostics, patient-centered acute care, and risk assessment methodologies.
- **The "X-Factor":** Dual-competency in Clinical Informatics, uniquely positioning me to contribute to hospital Quality Improvement (QI) initiatives, reduce medication errors, and integrate AI safely.

MEDICAL EDUCATION

Tbilisi State Medical University (TSMU) — Class of 2026

- **Degree:** Medical Doctor (MD) Candidate.
- **Clinical Rigor:** Recognized for academic excellence across multifaceted clinical rotations, with a strong emphasis on bedside triage and diagnostic reasoning.
- **Core Competency:** Extensive focus on disease pathophysiology, acute care management, and the integration of MedTech data architectures.

CORE CLINICAL COMPETENCIES

- Cardiovascular Risk Assessment
- Disease Pathophysiology
- Acute Care Triage (~50 Protocols)
- WHO HEARTS Protocol Implementation
- Vascular Surgery (Anastomosis)
- Advanced Suturing Techniques

MEDICAL RESEARCH & PUBLICATIONS

- **The Digital Transformation of Cardiology: A Narrative Review**
  Analyzed the integration of intelligent wearables in remote monitoring. Demonstrated how continuous data streams effectively shift cardiology toward proactive, personalized diagnostic models.
- **Cardiovascular & AI-Driven Diagnostics (Research in Progress)**
  Currently undergoing rigorous peer review. Directing extensive clinical trials and data analysis to validate the next generation of predictive cardiovascular diagnostics.
- **Cognitive Load Aware Evaluation of LLMs in Medical Diagnosis**
  Pioneered a novel framework evaluating AI within clinical settings. Prioritized the physician's mental experience to reduce cognitive load, solving a primary barrier to AI adoption in healthcare.

VERIFIED MEDICAL CERTIFICATIONS

- **World Health Organization (2026):** HEARTS Technical Package for Program Managers
- **GAAVS Association (2025):** Basics of Vascular Surgery: Mastery of Anastomosis Techniques
- **TSMU & IATROS (2025):** Basic Suturing: Principles and Techniques
- **MDPI (2026):** Official Delegate – 1st International Online Conference on Healthcare

CLINICAL INFORMATICS & QI PORTFOLIO

- *As a Full-Stack Developer, I architect decision-support systems designed specifically to **mitigate iatrogenic error, enhance diagnostic safety, and optimize hospital efficiency.** View live tools at Web-Project-Portfolio.*

CardioReference Ecosystem

- Engineered a strategic cardiology application utilized directly at the point of care.
- Integrates comprehensive disease pathophysiology with interactive ECG tools and predictive risk algorithms for rapid, data-driven cardiovascular assessments.

Safety-First Drug Calculator

- Built a precision calculation engine explicitly designed to mitigate human dosing errors.
- Incorporates specific patient biometrics (age, sex, weight, height) integrated with automated safety warnings and contraindication cross-checks.

Automated Lab Results Interpreter

- Pioneered an automated interpretation system for CBC, Thyroid, LFT, and KFT diagnostics.
- Streamlines workflow by automatically flagging abnormal values and suggesting clinical correlations, effectively minimizing manual data analysis errors.

EA Clinical Protocols Engine

- Digitized a symptom-based clinical protocol system covering ~50 common acute care presentations.
- Utilizes complex decision-tree logic to guide seamless assessment-to-management transitions.

SutureGuide Decision-Support

- Developed an intelligent algorithm evaluating wound location, tissue type, and tension.
- Outputs technical recommendations for optimal suturing materials and techniques.

VisionHealth & Vitality.Health

- Developed preventative health platforms providing interactive screening for 50+ visual disorders.
- Utilizes patient biometrics and fitness algorithms for personalized, data-driven nutritional planning.

INFORMATICS CERTIFICATIONS

- **SelectScience (2026):** Advancing Precision Oncology through a Digital-First Lab Model
- **SelectScience (2026):** Exploring an AI-Native Approach to Antibody Discovery
- **SelectScience (2026):** Modernizing Drug Discovery Informatics with AI and Machine Learning

TECHNICAL MATRIX

Full-Stack Web Dev	Calculation Engines	Clinical Data Architecture
Python & JavaScript	React.js & Node.js	SQL & NoSQL Architecture
AI/ML Integration	Cognitive Load Analytics	Clinical UI/UX Design